

Time Spent: 0 sec

Subject:

Biochemistry

QID: 400710

System:

Enzymes

The low-temperature-sensitive (LOT) genes are a group of genes in the yeast *Saccharomyces cerevisiae* that are expressed at higher levels as temperature decreases. These genes perform various functions within the cell, ranging from glucose metabolism to ribosome stabilization to RNA processing.

LOT6p is a homodimeric oxidoreductase enzyme that helps control oxidative stress by reducing a variety of quinones and related molecules. The active site of LOT6p contains a flavin mononucleotide (FMN) prosthetic group. In the first step of the LOT6p mechanism, a cofactor enters the active site and reduces the FMN to FMNH₂. The cofactor then leaves the active site, allowing the quinone substrate to enter. FMNH₂ then reduces the quinone and reverts to FMN in the process.

To characterize the kinetics of LOT6p, two experiments were conducted. In the first experiment, LOT6p was mixed with varying amounts of cofactor and a constant amount of the substrate 2,6-dichlorophenolindophenol (DCPIP), and the initial reaction rate for each cofactor concentration was plotted. In the second experiment, cofactor concentration was held constant while the DCPIP concentration was varied, and the reaction rate as a function of DCPIP concentration was recorded. Neither experiment showed any evidence of cooperativity.

A third experiment showed that the molecule resveratrol competitively inhibited LOT6p with respect to both substrates and that as the concentration of resveratrol was increased, the affinity of LOT6p for resveratrol decreased.

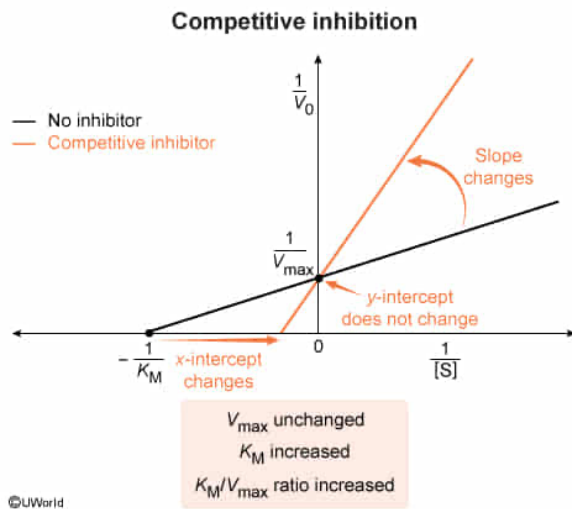
On a Lineweaver-Burk plot of LOT6p activity, which of the following is expected to change when resveratrol is added?

- ☐ A. The x- and y-intercepts, but not the slope [3%]
- ☐ B. The slope and y-intercept, but not the x-intercept [11%]
- ✓ ☒ C. The slope and x-intercept, but not the y-intercept [79%]
- ☐ D. The slope and both the x- and y-intercepts [4%]

Correct

79% answered correctly

Explanation:



A **Lineweaver-Burk plot** is a double-reciprocal plot in which $1/V_0$ is plotted as a function of $1/[S]$, where $[S]$ is substrate concentration and V_0 is the reaction rate at that substrate concentration. These plots are convenient for determining the enzyme kinetic parameters V_{max} and K_M because the plots are **linear**, with the **y-intercept representing $1/V_{max}$** and the **x-intercept representing $-1/K_M$** . In addition, the **slope of a Lineweaver-Burk plot equals the ratio K_M/V_{max}** .

Different types of inhibitors have different effects on V_{max} and K_M , and therefore have different effects on a Lineweaver-Burk plot. According to the passage, resveratrol is a **competitive inhibitor** of LOT6p. Competitive inhibitors cause an **increase in the K_M** value of an enzymatic reaction but **do not affect V_{max}** . Consequently, on a Lineweaver-Burk plot a competitive inhibitor is characterized by the fact that it **does not change the y-intercept ($1/V_{max}$)**. In contrast, the inhibitor **does cause a change in the x-intercept ($-1/K_M$)**, and consequently the **slope of the line (K_M/V_{max})**.

(Choice A) An inhibitor that changes the x- and y-intercepts but not the slope of a Lineweaver-Burk plot is an **uncompetitive inhibitor**. Uncompetitive inhibitors cause V_{max} and K_M to change by exactly the same factor (eg, if V_{max} decreases to half its original value, so does K_M), so the K_M/V_{max} ratio does not change.

(Choice B) An inhibitor that changes the slope and y-intercept but not the x-intercept of a Lineweaver-Burk plot is a **noncompetitive inhibitor**. Noncompetitive inhibitors are a special case of mixed inhibitor that bind free enzyme and enzyme-substrate complex with equal affinity and do not change K_M , so $-1/K_M$ also does not change.

(Choice D) An inhibitor that changes the slope, the x-intercept, *and* the y-intercept of a Lineweaver-Burk plot is a **mixed inhibitor** that binds both free enzyme and enzyme-substrate complex but with different affinities. These inhibitors decrease V_{max} and may either increase or decrease K_M but not by the same factor as V_{max} . Consequently, they

also change the $K_M/V_{\max}$ ratio.

Educational objective:

Lineweaver-Burk plots are linear plots of $1/V_0$ as a function of $1/[S]$. The y -intercept represents $1/V_{\max}$, the x -intercept represents $-1/K_M$, and the slope is equal to $K_M/V_{\max}$. Competitive inhibitors alter K_M but not $V_{\max}$, so they alter the slope and x -intercept but not the y -intercept of a Lineweaver-Burk plot.

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